Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Story, Michael D Ph.D.		
Project Title:	Gene Expression Profile Analysis As A Prognostic Indicator Of Normal Tissue Response To Simulated Space Radiations		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75235-7320	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	04/01/2003	**End Date:**	05/31/2004
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2004 per S. Krenek/JSC as well as PI final report submitted in 2005 (chgs made 8/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stivers, David N. (University of Texas MD Anderson Cancer Center)		
Grant/Contract No.:	NAG9-1522		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Individual differences in intrinsic radiosensitivity account for at least part of the observed variation in normal tissue responses to radiation. However, normal tissue responses to radiation exposure are also influenced by biochemical signaling associated with cell growth, proliferation, differentiation status, as well as other signaling events. In the past, no single assay or collection of assays could capture all of this information in an informative manner so as to predict normal tissue response. We hypothesize that new methodologies that examine global gene expression are capable of capturing this information and that this information can be used to predict normal tissue response, particularly late radiation sequelae. We will use microarray analysis to examine the underlying gene expression patterns of human skin cultures donated by patients treated for cancer by radiation. We have over skin fibroblast cultures from more than 100 individuals all with clinical response data. Furthermore, we have previously shown that in vitro SF2 values correlate with the patient's clinical normal tissue response. We will characterize the gene expression response of 20 skin fibroblast lines considered radiosenstive and 20 skin fibroblast lines considered normal after exposure to particulate radiations. Once molecular profiles are developed, a prognostic expression array will be created and tested against a test set of human skin fibroblasts from patients with known radiation response and SF2. Gene expression profiles determined after heavy particle exposure will also be compared to expression profiles currently being developed via another funding mechanism. This study should begin to allow for the characterization of individual risk for heavy particle radiation exposure, and perhaps help others to develop appropriate countermeasures to hazardous radiation exposures based upon gene expression patterns at the level of the individual astronaut.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study should begin to allow for the characterization of individual risk for heavy particle radiation exposure.
Task Progress:	This project was funded as a pilot project to determine the feasibility of using gene expression profiles to characterize the response of human cells to exposure to particulate radiations such as those encountered in the space flight environment. Because this was funded as a pilot project we focused our initial studies on logistics and appropriate experimental design, and then to test our hypothesis that there is a unique molecular response to specific particles, in this case, C and Fe, for primary human skin fibroblasts. The objectives of this project were to 1) determine molecular profiles (gene expression patterns) for normal human skin fibroblasts that have been exposed to heavy particles of increasing LET; 2) determine if there are unique, or signature, expression patterns that define exposure to a particular HZE particle. Primary human skin fibroblasts, our model, have been taken from patients being treated for either breast or head and neck cancer, and have been analyzed for low LET radiosensitivity and have been triaged into radiosensitive, normal, and radioresistant responses. Because this was a pilot project, we limited ourselves to 5 cell lines, all having normal response to radiation exposure. Our goals were two-fold. First, we wanted to develop survival plots to be able to determine RBE values for each particle in each cell line. Secondly, we wanted to develop expression signatures for HZE particle exposure. As this was a pilot project, examining the feasibility of both the biological model and the technology used, we feel that we have accomplished our goal. There is ample evidence provided that the models and technology can be applied to the Critical Path questions-28c, regarding how to estimate individual radiation carcinogenesis; 28d, regarding how to develop and validate effective biomarkers of the risk of space radiation on carcinogenesis; 30c, regarding how to estimate the latency period for degenerative tissue risks, following space radiation exposures-as well as others associated with radiation exposure. Furthermore, we have shown that intrinsic radiosensitivity, and potentially carcinogenesis, can be examined with this technology, that there are intrinsic differences among individuals, and that gene ontology analysis shows distinct differences in the signal transduction pathways that are radiation quality, and HZE particle and energy dependent.
Bibliography Type:	Description: (Last Updated: 02/12/2026)